

Q1 - More units and waves: Let's say we have a fat ribbon instead of a string vibrating. Instead of a mass per unit length, the ribbon has a mass per unit area σ . Instead of a tension, we now have a tension per unit length τ (if we have the same total force on a wider ribbon, the force on each bit will drop, and the wave speed will go down). What are the units of mass per area σ and force per length τ in kg, m, and s? What would the (approximate?) wave speed in the ribbon be in terms of σ and τ ?

Q2 - Repeating Middle C: What is the frequency of middle C (C4)? A4 (which is higher) has a frequency of 440 Hz.

Q3 - An ideal simple flute (with no holes) would have a lowest note where the wavelength is four times longer than the flute. If the lowest note of a flute is a middle C (261 Hz), how long should the flute be? The wave speed in the flute is the speed of sound. Your professor has finally realized that a real flute is not ideal because it has a big hole at the end where the flutist is blowing. In this case, the flute is open at both ends, and behaves like a string nailed at both ends. How long should the flute be now? Does this line up with the true value (about 610 mm, according to the web)?

Q4 - We saw that the wave speed on a violin's E-string is about 250 m/s. The piano's highest note is a C8 - 4 octaves above middle C. What frequency is this note? If the wave speed is about the same on a high piano string and a high violin string, how long would the piano string be? In practice, the piano strings are under a lot more tension and so the highest notes have a wave speed about double that of a violin.

Q5 - A piano has 88 keys. What is the ratio of the fundamental frequencies of the highest and lowest keys? If the vibrating length of the highest pitched string is 65mm, how long would the bottom string have to be if made of same wire under the same tension?